

Open-source in open-sky

Greenberg and O'Neel, heading the Portland State Aerospace Society (PSAS), plan to launch a Linux-powered rocket, equipped with an AMD 586 processor, to 55,000 feet at a speed three times faster than the speed of sound in September 2003. The rocket has an onboard computer with an AMD 586 processor, a Jumptec MOPS/520 PC/104+ board, along with a power supply, a PCMCIA card carrier for an 802.11b card to



transmit data to the ground and a carrier board for a 128 MB CompactFlash card for long-term storage. The flight computer runs a bare version of Debian Linux, with the 2.4.20 Linux kernel. Microcontrollers run everything from the inertial measurement system to the recovery device, which triggers a small explosive to deploy the parachutes. The flight computer transmits data to the ground through a 2.4 GHz wireless link.

snapshot

Business Process Outsourcing (BPO) market worldwide is expected to grow to \$122 billion in 2003

Source: Gartner, Aberdeen Group

Never say never again

In a total turnaround of their earlier policy, Metallica is now embracing the online world, and has just opened up www.metallicacvault.com, that comes with the tagline 'Download. Burn. Share. Kick A**'. Fans who buy Metallica's new album, will find a code inside the package that lets them download over six hours worth of content in the MP3 format, for free. Eventually, Metallica hopes to have several hundred hours of content on the site.

Three years ago, Metallica drummer Lars Ulrich and rapper Dr. Dre spearheaded the drive to bring down Napster, by compiling lists of over 3,00,000 users who were swapping their songs and other copyrighted material. Napster responded by



filing counter claims and then blocking material by specific artists, but eventually the service floundered.

Metallica released their new albums five days earlier than expected, claiming that "substandard versions" of the tracks were already available widely on the Internet.

After this announcement by Metallica, a host of artists are considering adding selected tracks online, to appease their fans. Roxio is restarting the controversial service—but it's going to be paid.



quoteworthy

"The most unfortunate thing is that India still seems to believe in proprietary solutions"

President A.P.J. Abdul Kalam praising open source software

"GameBoy is for 10-year-olds... If you're 20 or 25 years old, it's probably not a good idea to draw a GameBoy out of your pocket on a Friday night in a public place"

Ilkka Raiskinen, Nokia's head of entertainment and media, attacking Nintendo's Game Boy Advance

"...kids are just as susceptible to being bombarded by spam advertising inappropriate products and services."

Steve Cullen, Senior Vice President, Consumer and Client Product Delivery at Symantec. Their survey of netizens between the ages of 7 and 18 revealed that 47 per cent received e-mails with X-rated Web site links

tomorrow's technology

Memory a-billion

Nantero, a US based company has developed the first 10 GB nanotechnology memory (NRAM) device using standard chip production techniques. NRAM works with carbon nanotubes that are balanced on silicon ridges. Under differing electric charges, these tubes can be

placed into one of two positions, representing zero and one. Very less power is needed to place these tubes, being very small, and much more conductive than copper. This makes it easy to sense the position and read the data. Since they have a high tensile strength, they are expected to

survive over a trillion write cycles. Nantero solves the problem of misaligned nanotubes by using lithography and etching to remove them. The company is looking at a \$100 billion market of such NRAM chips that might eventually replace all forms of standard memory devices.

■ Linus Torvalds is taking a long leave of absence from Transmeta, and is going to be working in the Open Source Development Lab fulltime.